BCS-601	COMPUTER NETWORK	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	Study the basic taxonomy and terminology of the computer networking and enumerate the layers of OSI model and TCP/IP model.
CO2	Acquire knowledge of Application layer and Presentation layer paradigms and protocols.
CO3	Study Session layer design issues, Transport layer services, and protocols.
CO4	Gain core knowledge of Network layer routing protocols and IP addressing.
CO5	Study data link layer concepts, design issues, and protocols.

MODULE-I

Introduction Concepts: Goals and Applications of Networks, Network architecture, The OSI reference model, TCP/IP model ,services, Network Topology Design - Delay Analysis, Back Bone Design, Local Access Network Design, Physical Layer Transmission Media, Switching methods.Data Link Layer - Elementary Data Link Protocols, Sliding Window protocols, Error Handling.

MODULE-II

Medium Access sub layer - Channel Allocations, Medium Access protocols, Ethernet, Wireless LANS.Network Layer: Network Layer - Point - to Pont Networks, routing, Congestion control, IP packet, IP address, IPv6.Transport Layer: Transport Layer – Design issues, connection management.

MODULE-III

Session Layer- Design issues, remote procedure call. Presentation Layer-Design issues, Data compression techniques, cryptography – Symmetric-Key and Asymmetric –Key, Security services.Application Layer: DNS, File Transfer Access, Electronic mail, SNMP.

Text Books:

1. Forouzen, "Data Communication and Networking", TMH
2. A.S. Tanenbaum, Computer Networks, Pearson Education

Reference Books:

1. W. Stallings, Data and Computer Communication, Macmillan Press
2. AnuranjanMisra, “Computer Networks”, Acme Learning
3. G. Shanmugarathinam, ”Essential of TCP/ IP”, Firewall Media

Course Outcomes: After the completion of the course the student will be able to:

CO1	Describe the functions of each layer in OSI and TCP/IP model.
CO2	Explain the functions of Application layer and Presentation layer paradigms and Protocols.
CO3	Describe the Session layer design issues and Transport layer services.
CO4	Classify the routing protocols and analyze how to assign the IP addresses for the given network.
CO5	Describe the functions of data link layer and explain the protocols.